

Server Bootstrap

server.ts

1. Load .env via dotenv
2. Validate config (Zod schema)
3. Connect MongoDB Atlas
4. Register DI container (repos, services, controllers)
5. Create Express app
6. Listen on PORT

Middleware Pipeline

helmet

cors

rateLimit

body parsers

morgan

Route Dispatch

Route Matcher

/api/v1/*

no auth

authMiddleware

Public Routes

GET /health → liveness
POST /auth/* → register,
login, refresh, OTP
POST /diagnose → diagnosis

Protected Routes

JWT

/users → profile CRUD
/crops → crop management
/scans → scan records
/tasks, /notes, /advisories

Controller Layer

— validate request body → invoke service method → shape JSON response

Processing Chain

Service

business rules

DTO

Repository

data access

ODM

MongoDB

document store

catch

Error Handler

ApiError → JSON

JSON Response

structured envelope

Design rationale — Middleware executes in strict registration order. Public routes (/health, /diagnose, /auth/*) bypass JWT verification; protected routes inject req.user via authMiddleware. Controller → Service → Repository → MongoDB enforces separation of validation, business logic, and persistence.